

6.2. 29
29.

U 和 V 为正交矩阵

$$\text{22) } U^{-1} = U^T \quad V^{-1} = V^T$$

则 UV 可逆.

$$\therefore (U \cdot V) \cdot (U \cdot V)^T$$

$$= UV \cdot V^T \cdot U^T$$

$$= i$$

$$\therefore (UV)^{-1} = (UV)^T$$

∴ UV 也是正交矩阵

6.3 13. 14. 22

$$13 \quad \hat{z} = \frac{z \cdot v_1}{v_1 \cdot v_1} \cdot v_1 + \frac{z \cdot v_2}{v_2 \cdot v_2} \cdot v_2$$

$$= \frac{2}{3} v_1 - \frac{7}{3} v_2$$

$$= \begin{bmatrix} -1 \\ -3 \\ -2 \\ 3 \end{bmatrix}$$

$$14 \quad \hat{z} = \frac{z \cdot v_1}{v_1 \cdot v_1} \cdot v_1 + \frac{z \cdot v_2}{v_2 \cdot v_2} \cdot v_2$$

$$= \frac{1}{2} v_1 + 0 v_2$$

$$= \begin{bmatrix} 1 \\ 0 \\ -\frac{1}{2} \\ 3 \\ -\frac{3}{2} \end{bmatrix}$$

22

a. ✓

b. ✓

c. ✓

d. x 是 proj_W

e. x 若 $n > p$, 则 $\{v_i\}$ 不构成 $\mathbb{R}^n$, 不成基.

6.5 12, 17, 18, 19

12

∵ A 中 a_1, a_2, a_3 相互正交

$$\therefore \hat{b} = \frac{b \cdot a_1}{a_1 \cdot a_1} a_1 + \frac{b \cdot a_2}{a_2 \cdot a_2} a_2 + \frac{b \cdot a_3}{a_3 \cdot a_3} a_3$$

$$= \frac{1}{3} \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \end{bmatrix} + \frac{14}{3} \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \end{bmatrix} - \frac{5}{3} \begin{bmatrix} 0 \\ -1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \\ 3 \\ 6 \end{bmatrix}$$

∵ x 是 A 中列表示 b 的权

$$\therefore x = \begin{bmatrix} \frac{1}{3} \\ \frac{14}{3} \\ \frac{5}{3} \end{bmatrix}$$

17 a. ✓

b. ✓

c. x 符号反了.

d. ✓

e. ✓

18 a. ✓

b. ✓

c. ✓

d. x A 的列必须线性无关.

e. x 有可能是病态的 fx 计算 $Rx = Q^T b$ 更快

$$19. Ax = 0$$

$$A^T(Ax) = A^T \cdot 0 = 0$$

$$\text{若 } A^T Ax = 0 \text{ 则 } x^T (A^T A x)$$

$$= x^T \cdot 0 = 0$$

$$\therefore x^T A^T A x = 0$$

$$\therefore x^T A^T A x = (Ax)^T Ax = 0$$

$$\|Ax\| = 0 \text{ 则必有 } Ax = 0$$

6.4.13

13 由例4

$$R = Q^T \cdot A = \begin{bmatrix} \frac{5}{6} & \frac{1}{6} & -\frac{3}{6} & \frac{1}{6} \\ -\frac{1}{6} & \frac{5}{6} & \frac{1}{6} & \frac{3}{6} \end{bmatrix} \begin{bmatrix} 5 & 9 \\ 1 & 7 \\ -3 & -5 \\ 1 & 5 \end{bmatrix} = \begin{bmatrix} 6 & 12 \\ 0 & 6 \end{bmatrix}$$

验证:

$$QR = \begin{bmatrix} \frac{5}{6} & \frac{-1}{6} \\ \frac{1}{6} & \frac{5}{6} \\ \frac{-3}{6} & \frac{1}{6} \\ \frac{1}{6} & \frac{3}{6} \end{bmatrix} \cdot \begin{bmatrix} 6 & 12 \\ 0 & 6 \end{bmatrix} = \begin{bmatrix} 5 & 9 \\ 1 & 7 \\ -3 & -5 \\ 1 & 5 \end{bmatrix} = A$$

∴ 结果正确

7.1 19 22

9. 若 $\lambda = 7$, 求得基为 $\left\{ \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$

转换为正交基: $\left\{ \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \\ 5 \end{bmatrix} \right\}$

单位化: $\begin{bmatrix} -1/\sqrt{5} \\ 2/\sqrt{5} \\ 0 \end{bmatrix} \begin{bmatrix} 4/\sqrt{45} \\ 2/\sqrt{45} \\ 5/\sqrt{45} \end{bmatrix}$

若 $\lambda = -2$ 基为 $\begin{bmatrix} -2 \\ -1 \\ 2 \end{bmatrix}$ 单位化 $\begin{bmatrix} -2/3 \\ -1/3 \\ 2/3 \end{bmatrix}$

$$\therefore P = \begin{bmatrix} -1/\sqrt{5} & 4/\sqrt{45} & -2/3 \\ 2/\sqrt{5} & 2/\sqrt{45} & -1/3 \\ 0 & 5/\sqrt{45} & 2/3 \end{bmatrix}$$

$$D = \begin{bmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

$$\det(A - \lambda I)$$

$$\begin{array}{ccc|c} 2-\lambda & -1 & 1 & 0 \\ -1 & 2-\lambda & -1 & 0 \\ 1 & -1 & 2-\lambda & 0 \end{array}$$

$$= (2-\lambda)((2-\lambda)^2 - 1) + (\lambda - 2 + 1) + 1 + \lambda - 2,$$

$$= (2-\lambda)^3 - (2-\lambda) + 2(\lambda - 2) + 2$$

$$= (2-\lambda)^3 - 3(2-\lambda) + 2$$

$$2-\lambda = 1 \quad \lambda = 1 \text{ 重数 } 2$$

$$2-\lambda = -2 \quad \lambda = 4$$

若 $\lambda = 1$ 求基: $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ 单位化

$$\begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} \begin{bmatrix} -1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}$$

若 $\lambda = 4$ 求基: $\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ 单位化 $\begin{bmatrix} 1/\sqrt{3} \\ -1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$

$$\therefore P = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 1/\sqrt{3} \\ 1/\sqrt{2} & 0 & -1/\sqrt{3} \\ 0 & 1/\sqrt{2} & 1/\sqrt{3} \end{bmatrix} \quad D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$